Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

Machine Learning and AI Engineer with **3 years** of industry experience taking **ML and GenAI systems** from **prototype to production** in a regulated financial services setting: **Python** services and scheduled batch workloads, containerised with **Docker** and deployed on **AWS** through a one-click **Jenkins and GitLab CI/CD pipeline**. Built and operated production **data pipelines** processing over **300 documents a day**, designed a **scoring and ranking algorithm** and completed a **multi-architecture deep learning study** for an **MSc in Data Science (University of Bristol)**, dissertation with **AstraZeneca**. Acted as **de facto team lead for up to 4 interns** and was twice recognised with **Best Team Project** (2024, 2025). Seeking to build the training, deployment and monitoring systems behind Checkout.com's ML-driven dispute optimisation suite.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Production ML, GenAI & Real-Time Services

- Designed and built **production machine learning and GenAI applications** end to end (**ReactJS** frontend, **Python FastAPI** microservices, retrieval with **LlamaIndex** and **Azure OpenAI** GPT-3.5 & Llama-3) serving internal users across **Compliance, Client Service and Investment**, with a separate **pgvector** database and **Azure OpenAI embeddings** per department.
- Built **agentic tool-calling** where AI agents call **Python functions** as tools to run **SQL-backed fund comparisons** on our **PostgreSQL** data and generate **PDF reports** on request; versioned every **prompt change in Git** and had the **Compliance** department review each release.
- Took applications from **prototype to real users** and iterated on what broke; allocated models by cost, **self-hosting Llama-3 on a GPU EC2 instance** for the largest department and keeping the stronger **Azure OpenAI** outputs for decision makers.

MLOps, Deployment & Production Reliability

- Designed the infrastructure architecture for internal analytics platforms and GenAI applications on **AWS** (**EC2 including a GPU instance, S3, ECR and Lambda**); packaged the **FastAPI** microservices (uvicorn) as **Docker** containers and automated deployment of the separate frontend and backend repositories (dev and prod) through a one-click **Jenkins and GitLab CI/CD pipeline**.
- Operated both **real-time and batch workloads**: user-facing services on FastAPI microservices, and a daily rating-email pipeline run on **AWS SageMaker** and scheduled by an **EventBridge** rule that triggers a **Lambda** function at 9 AM; troubleshot and improved the pipeline when a source changed its email format.
- Implemented **RBAC** with a Python authentication function against the company **LDAP** server; restricted access to company private IPs using **AWS Security Groups**; deployed an **Nginx** reverse proxy, one of the fixes that took the applications from repeatedly failing to passing the security team's **VAPT**.

Data Pipelines, Backend Services & Stakeholder Delivery

- Designed a **Normalized SQL Schema** (**PostgreSQL**) and wrote **Python** ingestion scripts for the **Morningstar API** covering every fund in the Indian mutual fund market (2023–2025), refreshed fortnightly and monthly and integrated with legacy analyst data; developed **Optimized SQL queries, Functions and Procedures**.
- Built a **Python** extraction pipeline that parses over **300 daily emails** from credit rating agencies into structured data, fuzzy-matches them to tracked companies and sends analysts a daily summary, saving **60 analyst hours a month**; replaced **AWS Textract** with **Azure OpenAI** when a rating agency began sending emails as images.
- Analyzed customer data using **Python Pandas** and **Market Basket Analysis (Apriori Algorithm)**; migrated financial analysts from Excel to **Power BI** dashboards, improving efficiency by **10 hours a week**.
- Worked directly with stakeholders in **Client Service, Compliance and Investment** to turn requirements into working tools; acted as **de facto team lead for up to 4 interns** in my final year; recognised with **Best Team Project** twice (2024 rating-email automation, 2025 department RAG chatbot).

PROJECTS

- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built. Data preprocessing pipeline, scoring algorithm and RAG-based AI assistant for a tool, built as FastAPI microservices with a React frontend, that ranks cancer cell lines by target inclusion and exclusion genes; validated and refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **Wingman:** AI dating-coach web app with 108 LLM personas across 12 archetypes and a vision model scoring uploaded photos (ReactJS, Python FastAPI, Kimi K2 and DeepSeek V4 Pro via NVIDIA NIM). LLM output treated as untrusted: validated with Pydantic, repaired and cached by input hash; rate limits shaped the design (12 calls instead of 108, serialised calls, results streamed to the UI). GitHub: https://github.com/Arishtanemi4/wingman.git
- **University of Bristol MSc Applied Data Science Group Project:** VisionCaptioner (Team project. Architecture study comparing CNN/ViT/ResNet encoders and LSTM/attention decoders for generating image descriptions, benchmarked against BLIP/SigLIP and commercial VLMs.) GitHub: https://github.com/Arishtanemi4/ata-team30.git
- **B.Tech Final Year Project:** Udacity self-driving car simulator. (A self-driving car trained using a **Convolution Neural Network (CNN)** model.)

ACHIEVEMENTS

- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)

SKILLS

- **Tools:** Docker, Jenkins, GitLab (CI/CD), AWS (EC2 including a GPU instance, S3, ECR, Lambda, EventBridge, SageMaker), Azure OpenAI, LlamaIndex, Nginx, Linux, LDAP, Power BI, VS Code, Claude Code.
- **Languages:** Python (pandas, scikit-learn, FastAPI, Pydantic, uvicorn), SQL, R (dplyr, tidyverse, ggplot2), NodeJS (express, ReactJS), Shell
- **Databases:** PostgreSQL (pgvector, SQL functions and procedures), SQLite